Power & Protection

文件: 01_power.kicad_sch

MCU & Debug

文件: 02_mcu.kicad_sch

IMU

文件: 03_imu.kicad_sch

TTL Bus AUTO

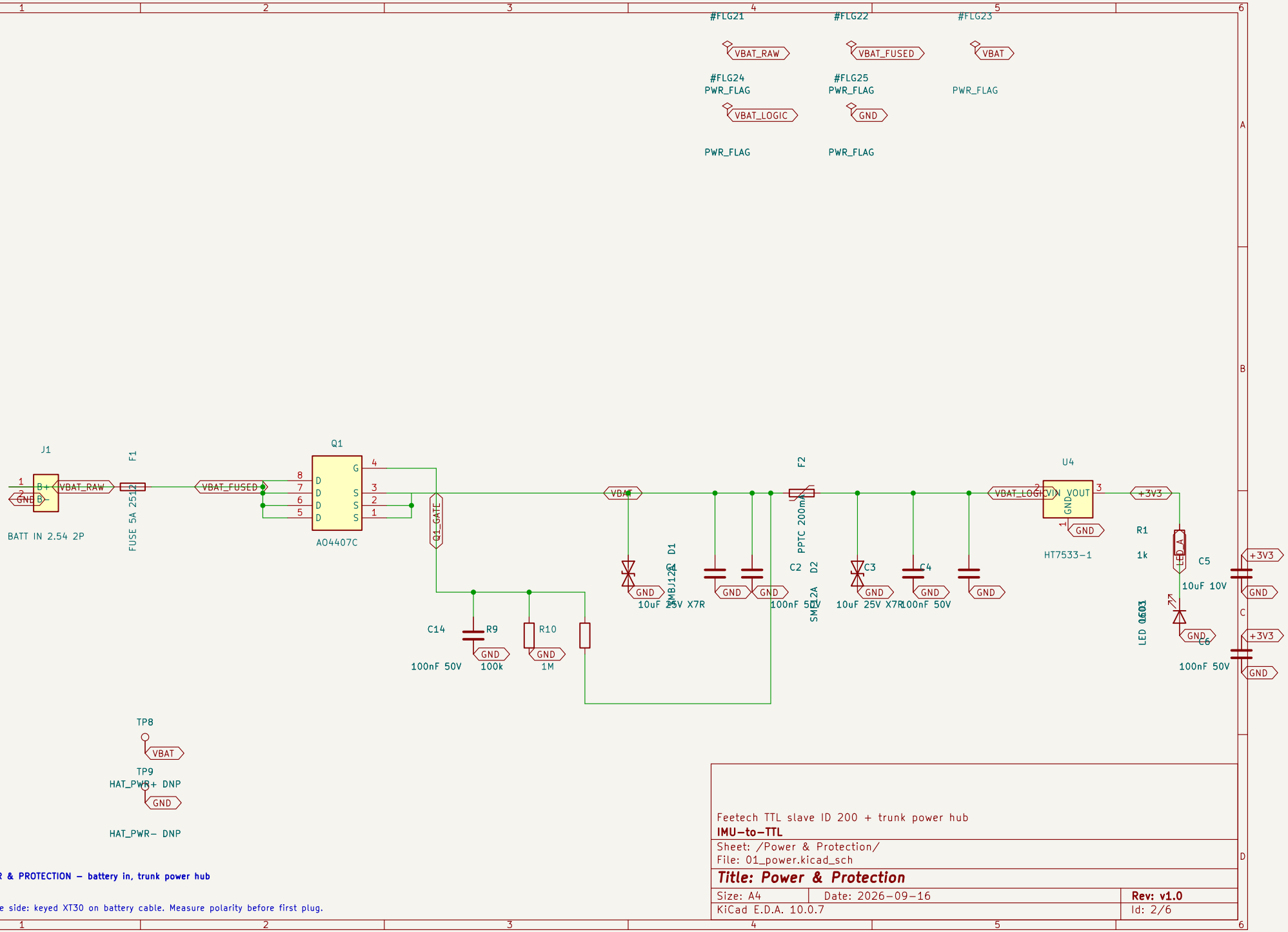
Feetech TTL slave ID 200 + trunk power hub for MicroDuck HD-1910 build

IMU-to-TTL

Sheet: /
File: imu_to_ttl.kicad_sch

Title: IMU-to-TTL board

Size: A4	Date: 2026-09-16	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 1/6



POWER & PROTECTION — battery in, trunk power hub

J1 wire side: keyed XT30 on battery cable. Measure polarity before first plug.

Q1: D=VBAT_FUSED S=VBAT G via 100k/1M + 100nF -> reverse protection with ~10ms soft start.

Per-branch current budget: PH 2.0 contact = 2A. J_NECK also feeds the head HAT.

Feetech TTL slave ID 200 + trunk power hub

IMU-to-TTL

Sheet: /Power & Protection/

File: 01_power.kicad_sch

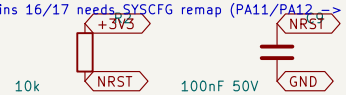
Title: Power & Protection

Size: A4	Date: 2026-09-16	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 2/6

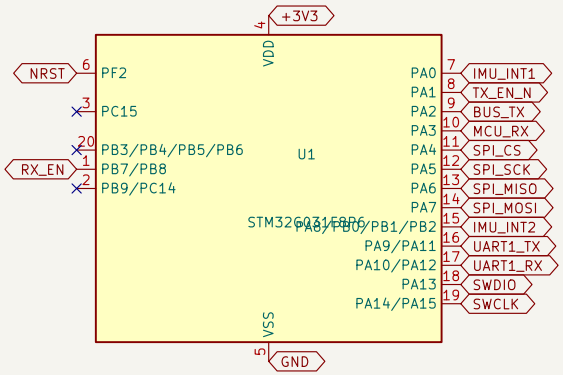
MCU — STM32G031F8P6 (TSSOP-20)

Clock: HSI16 only. No crystal — PC14 is LSE, NOT HSE (initial-review fix).

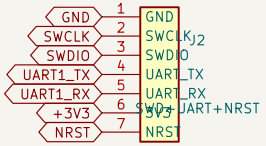
USART1 debug on pins 16/17 needs SYSCFG remap (PA11/PA12 -> PA9/PA10).



Unused pins: default analog high-Z after reset; left unconnected by design.





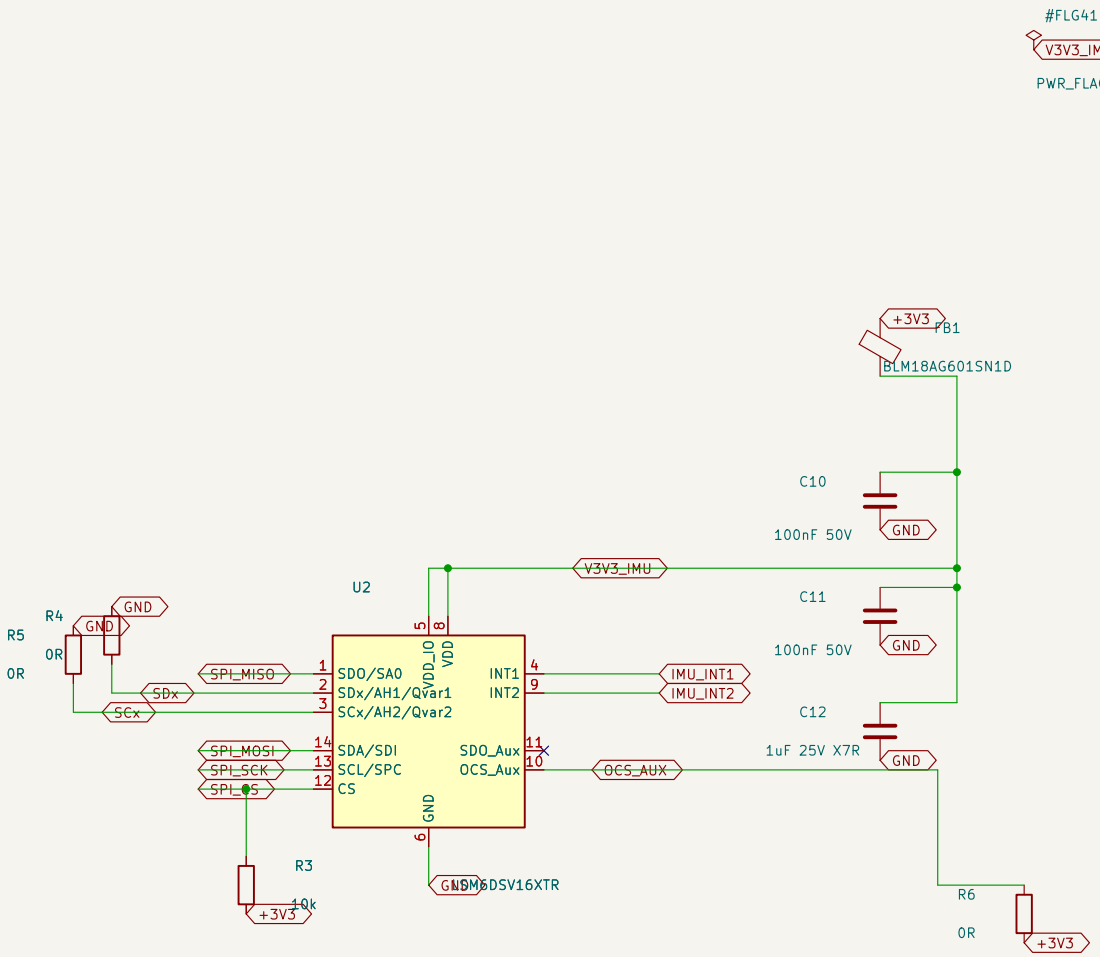


Feetech TTL slave ID 200 + trunk power hub
IMU-to-TTL

Sheet: /MCU & Debug/
File: 02_mcu.kicad_sch

Title: MCU & Debug

Size: A4	Date: 2026-09-16	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 3/6



IMU — LSM6DSV16X (SPI + SFLP)

Gyro full scale must be +/−500dps to match duck-control decode.

Mount axis: trunk = [+raw_z, +raw_y, −raw_x]; silkscreen X/Y/Z on board.

SDO_Aux unused -> NC (marked). SDx/SCx grounded via 0R (cuttable).

SDx/SCx pull-downs (R4/R5) terminate the aux I2C bus; OCS_AUX pull-up R6 per datasheet.

Feetech TTL slave ID 200 + trunk power hub

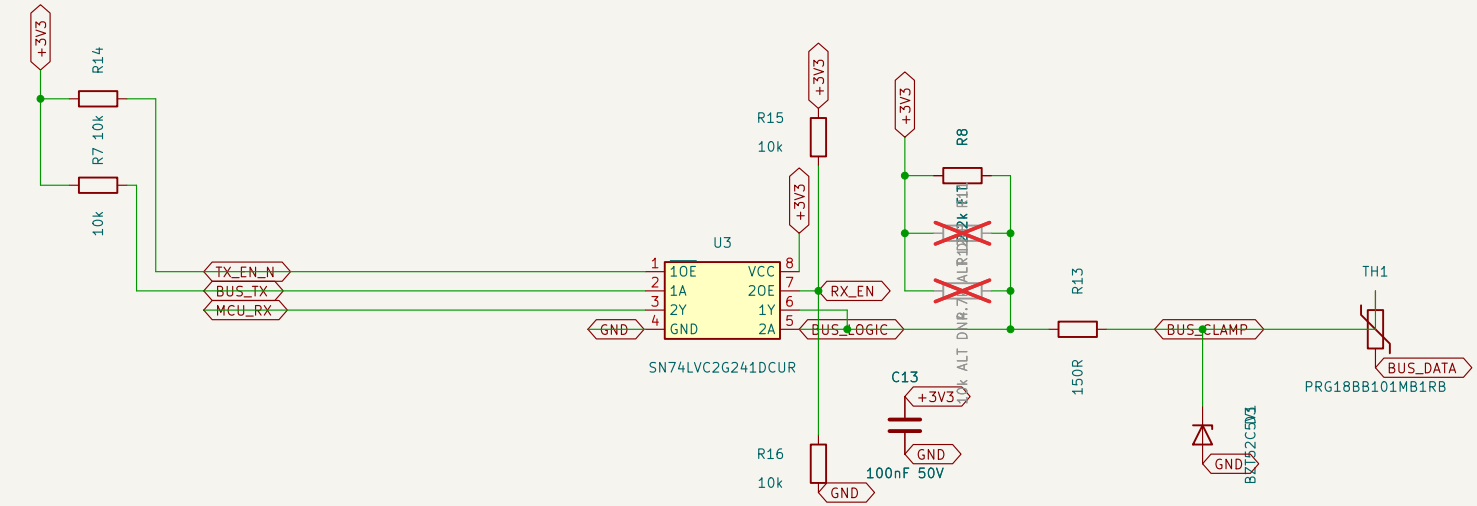
IMU-to-TTL

Sheet: /IMU/

File: 03_imu.kicad_sch

Title: IMU

Size: A4	Date: 2026-09-16	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 4/6



TTL BUS v2 — 2G241: PA1=TX_EN_N(1OE#) + PB7=RX_EN(2OE) + R16 boot strap (matches real board)

PA3 RX is direct on BUS_LOGIC: FT 5V-tolerant pin + 150R + 5.1V zener + 100R PTC.

Echo: during TX the UART sees its own bytes on RX — firmware must ignore RX while transmitting.

1Mbps timing gate: SPICE sweep (2.2k/4.7k/10k x 100/200/300pF) before PCB.

R8/R11/R12 = BUS_LOGIC pull-up options, fit ONE (2.2k default / 4.7k / 10k for trunk length)

Feetech TTL slave ID 200 + trunk power hub

IMU-to-TTL

Sheet: /TTL Bus AUTO/
File: 04_bus.kicad_sch

Title: TTL Bus AUTO

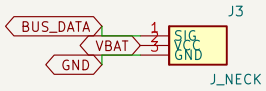
Size: A4	Date: 2026-09-16	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 5/6

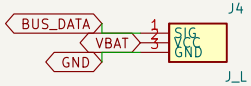
CONNECTORS — Feetech PH 2.0, all same pinout

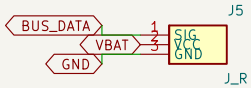
PINOUT: 1=SIG(TTL) 2=VCC(bus) 3=GND — HD-1910 spec A/0. NOT Dynamixel order!

All three bus ports are one shared net; VDD/GND pass through with wide 2oz copper.

Per-port current <= 2A (PH contact). J_NECK also carries the head HAT input.



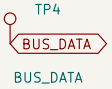




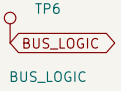














Feetech TTL slave ID 200 + trunk power hub

IMU-to-TTL

Sheet: /Connectors & Test/
File: 05_connectors.kicad_sch

Title: Connectors & Test

Size: A4	Date: 2026-09-16	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 6/6